

EASTERN INTERNATIONAL UNIVERSITY

BECAMEX BUSINESS SCHOOL



COURSE: SCLM 479 INTEGRATED LOGISTICS

AND SUPPLY MANAGEMENT

Strategic Purchasing & Supply Chain Case Study:

Rebisco Vietnam (Vision 2026–2032)



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CHAPTER 1: FORECASTING

I. Executive Summary

This paper outlines a demand forecasting project where we will develop several quantitative models to generate monthly demand forecasts for individual Stock Keeping Units (SKUs) over a 72-month period. The primary purpose is to help the business make better, data-driven decisions. Our group is using Python and Machine Learning techniques to explore the data and develop these predictive models.

II. Forecast Scope & Parameters

1. **Forecast Horizon:** 6 Years (January 2027 – December 2032).
2. **Data Granularity:** SKU Level (Bottom-up approach).
3. **Time Bucket:** Monthly intervals to capture seasonality and cyclical trends.

III. Data Understanding

1. Data overview

The dataset contains 1,656 records, giving a detailed picture of Rebisco's sales history. Each record captures important information that helps us understand demand patterns:

Time information: Year and Month.

Product details: Brand, Product Name, and Pack Size.

Sales volume: Measured in Metric Tons (MT).

The data covers a six-year period from 2021 to 2026, including 15 different SKUs across five brands such as *Doowee Donut*, *KrimStix*, *Rebisco Crackers*, *Rebisco Extreme*, and *Whoopie*. This level of detail is valuable because it allows us to see:

Seasonality: Some products sell more in certain months and less in others, often linked to holidays or special times of the year.

Trends: The overall direction of sales over several years, showing whether demand is going up, going down, or staying steady.

Product lifecycle: The journey of each product, from when it first appears in the market, to when it grows, reaches its peak, and eventually slows down.

By examining sales at the product level, we can see both the overall brand performance and the specific sales patterns of each product. This makes our forecasts more accurate and helps the company make supply chain decisions with greater confidence.

2. Data Preprocessing & Pipeline Construction Total Demand over Time

To move from raw sales records to a forecasting model, we first prepared the dataset using Python. Python was chosen because it has powerful libraries that make it easier to clean, organize, and analyze large amounts of data. The goal was to make sure the data was accurate, consistent, and ready for forecasting. We followed four key steps:

2.1. Numerical Normalization & Type Casting

In the raw dataset, the sales volume values were written with commas instead of decimal points. For example, a number like 174,31 appeared instead of 174.31. This caused a problem because Python would read these values as text rather than numbers. To fix this, we used Python to first convert the column into text, replace the commas with periods, and then change the entire column into a numeric format called float64. After this step, the data could be recognized as proper numbers. This was important because Python's libraries, such as NumPy and Pandas, can only perform calculations like adding, averaging, or forecasting when the data is numeric. Without this correction, the sales volumes could not be aggregated or analyzed correctly.

2.2. Temporal Feature Engineering

In the original dataset, the time information was split into two separate columns, one for the year and one for the month. This setup made it harder to see the data as a continuous timeline. To make things clearer, we used Python to combine the two into a single "Date" column, written in the standard format *YYYY-MM-DD*.

After creating the new date column, we used Python's Pandas library to recognize it as real dates instead of just text. This gave the dataset a proper timeline, allowing Python to follow the order of months and years correctly. With this change, it became much easier to see meaningful patterns, like which months consistently have higher sales, how demand shifts from year to year, and where seasonal peaks or slow periods occur.

2.3. Logical Sorting & Temporal Order

After creating the date column, the next step was to make sure the data followed the right timeline. Forecasting models, like SARIMA or Prophet, learn by looking at past sales and then predicting what comes next. For that reason, the data has to be arranged in the correct order.

We organized the dataset so that sales began with the first entries in 2021 and continued step by step until the end of 2026. This ordering is essential because if the records were out of place, for example, showing 2024 before 2022, the model could misinterpret the information and

produce inaccurate results. By keeping the data in proper order, the dataset presents a clear timeline of demand, showing how sales changed over the years and making it reliable for forecasting.

2.4. Aggregation & Frequency Alignment

Because we want to forecast sales at a monthly level, we combined the data from all 15 products (SKUs) into one total number for each month. To do this, we grouped the records by the new date column and added up the sales volumes.

Next, we set the dataset to follow a regular monthly timeline using a “Month Start” setting (`asfreq('MS')`). This makes sure every month is included in the data. If a month had no sales, it still shows up as an empty entry instead of being skipped. Keeping these zero-sales months visible is important because forecasting models need a complete timeline to correctly spot seasonal patterns and changes in demand.

3. Exploring Data Analysis (EDA)

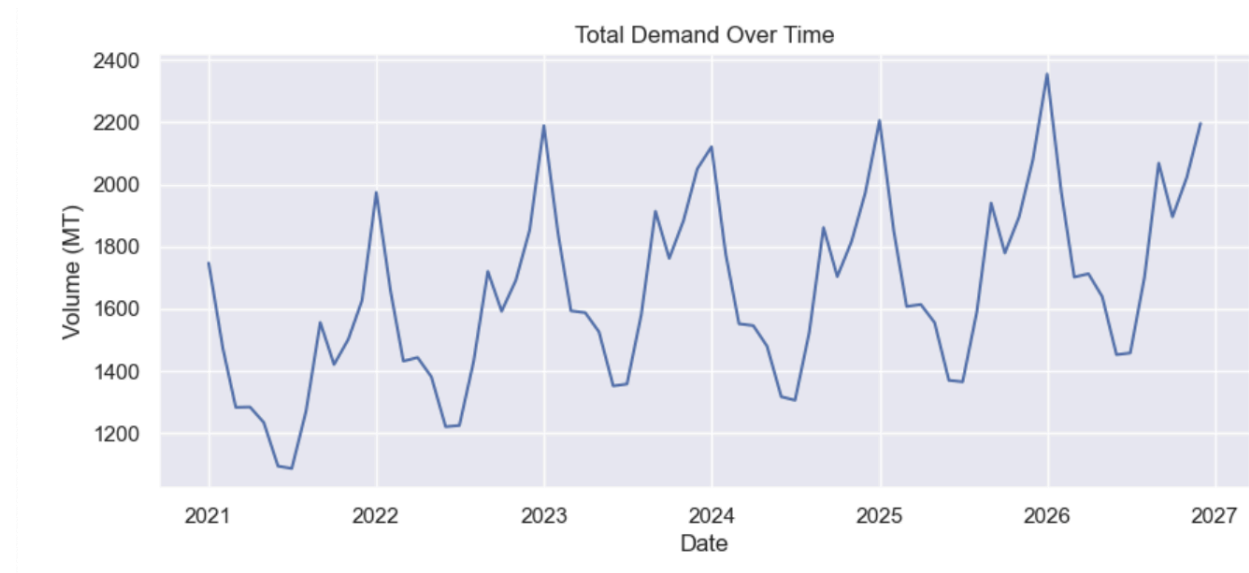


Figure 1: A time-series line chart showing total demand over time

Figure 1 illustrates the total demand of Rebisco Vietnam over the period 2021–2026. Based on the computed demand data, total annual demand increased from 16,576.58 MT in 2021 to 22,174.28 MT in 2026, reflecting a clear upward trend in overall demand. This growth indicates an expanding market and increasing consumption of Rebisco products.

In addition to the long-term trend, the demand series exhibits recurrent fluctuations at similar points each year, indicating the presence of seasonality. Such a pattern is consistent with a multiplicative time-series structure, expressed as:

$$Y_t = T_t \times S_t \times E_t$$

where Y_t represents observed demand, T_t the trend component, S_t the seasonal component, and E_t the random error term.

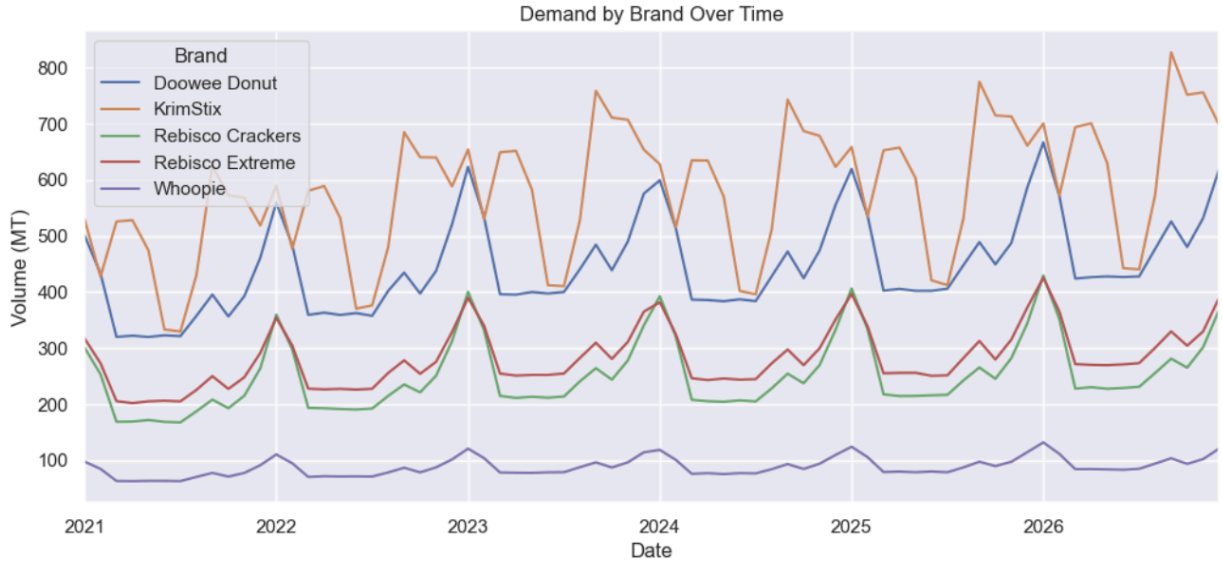


Figure 2: Comparative time-series line

Presents demand trends at the brand level. Aggregated brand data show that while demand magnitudes differ across brands, all brands follow a similar upward trajectory over time and exhibit comparable seasonal movements. This suggests that brand demand is influenced by common market-wide seasonal factors, while differences in volume are primarily driven by brand-specific demand levels.

Accordingly, brand-level demand can be represented as:

$$Y_{i,t} = T_{i,t} \times S_t \times E_{i,t}.$$

where the seasonal component S_t is shared across brands.

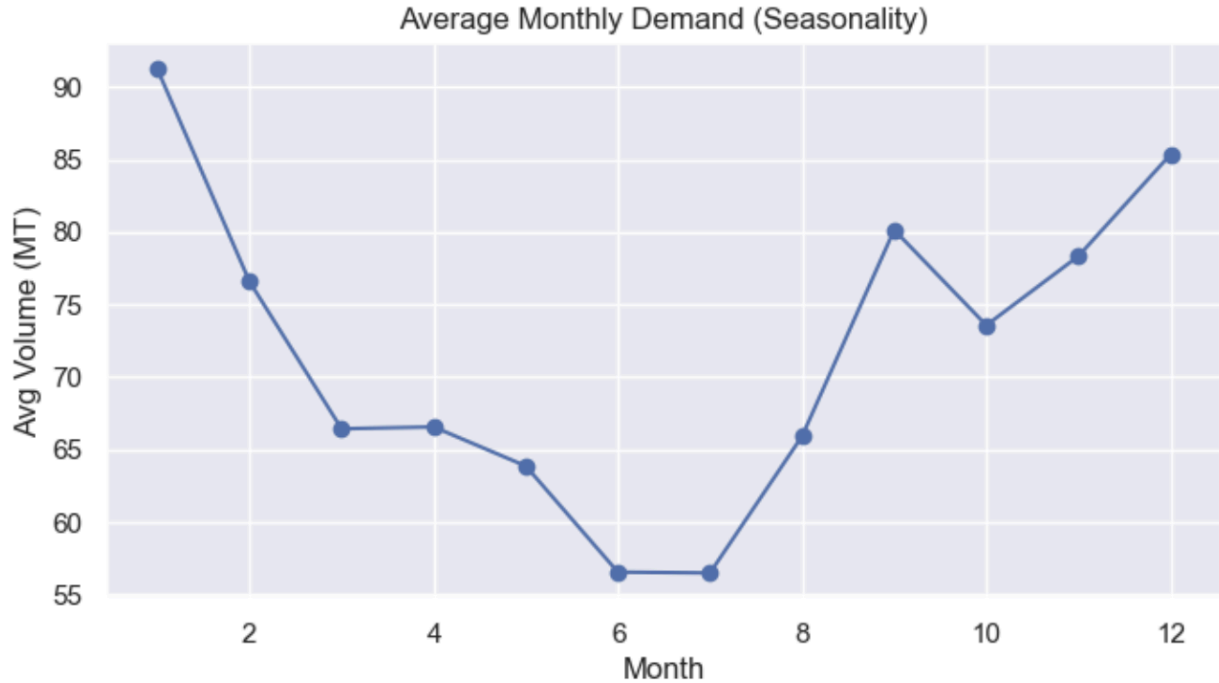


Figure 3: Average Monthly Demand each year from 2021 to 2026

Average Monthly Demand (Seasonality) shows the average monthly demand computed from historical data. Demand is highest in January (91.23 MT) and December (85.34 MT), while the lowest demand is observed in June (56.57 MT) and July (56.51 MT). This consistent monthly variation confirms the existence of a strong seasonal pattern.

Seasonality is quantified using the seasonal index:

$$S_m = \frac{\text{Average demand in month } m}{\text{Overall average demand}}$$

The empirical evidence from the Excel data confirms that Rebisco's demand exhibits both a positive trend and stable seasonality. Therefore, the Holt–Winters multiplicative model is appropriate for average demand in subsequent periods.

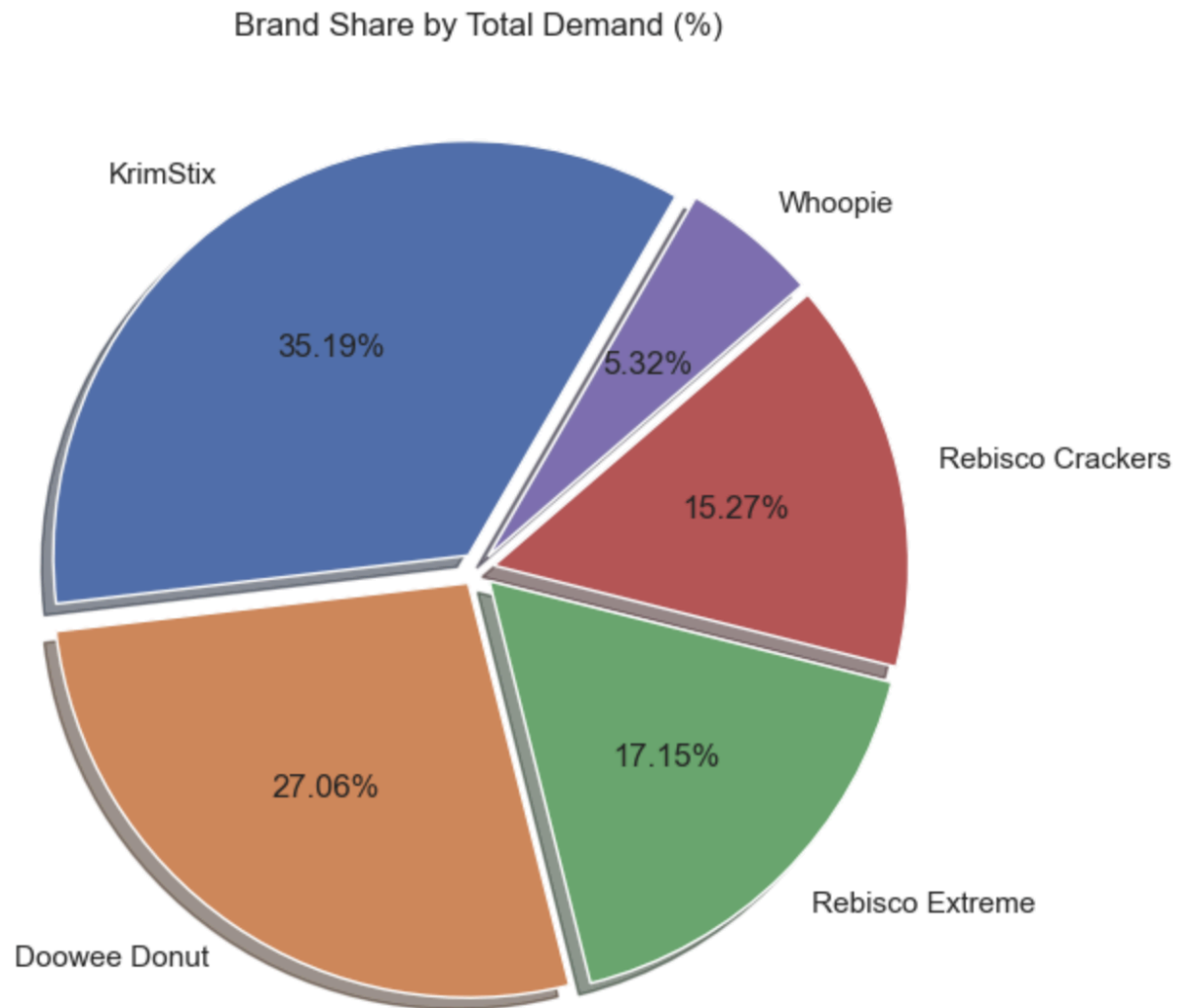


Figure 4: Brand Share by Total Demand (%)

The pie chart illustrates the distribution of total demand by brand over the observed period. KrimStix is the dominant brand, accounting for 35.19% of total demand, followed by Doowee Donut with 27.06%. Together, these two brands contribute more than 62% of total demand, indicating a high level of demand concentration. Rebisco Extreme and Rebisco Crackers represent moderate shares at 17.15% and 15.27%, respectively, while Whoopie contributes only 5.32%, reflecting its niche market position.

This distribution suggests that overall demand performance is largely driven by a small number of core brands.

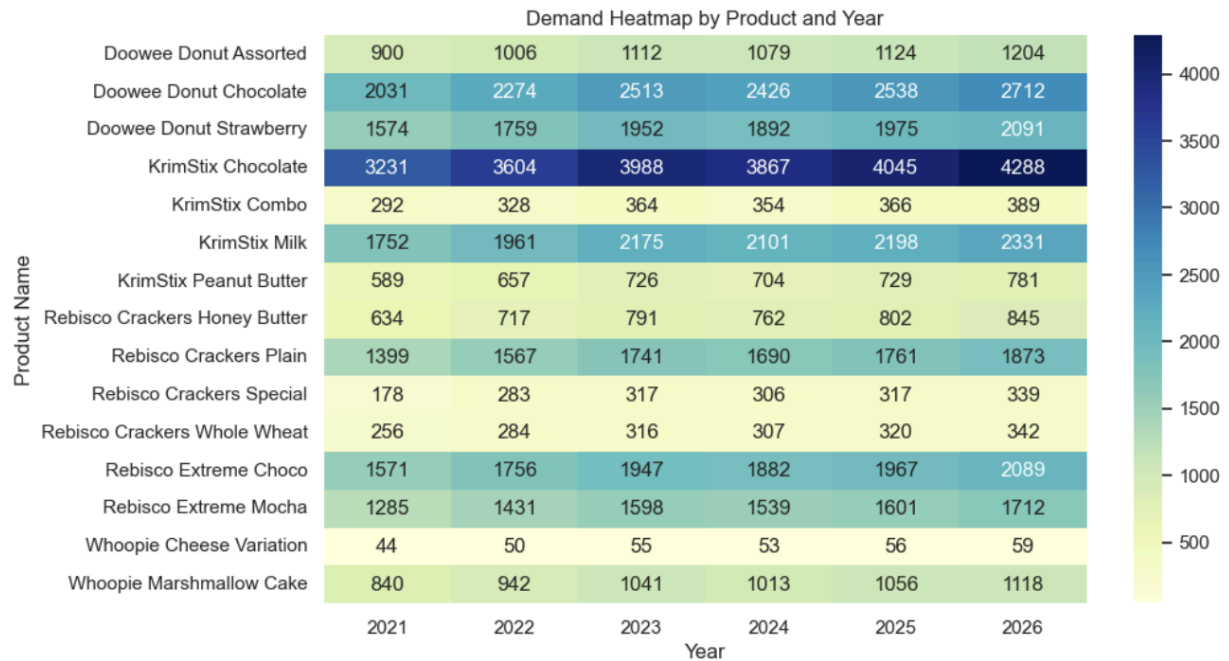


Figure 5: Matrix Heatmap of Demand

The demand heatmap provides a detailed view of SKU-level demand patterns from 2021 to 2026. High-volume SKUs such as KrimStix Chocolate and Dowee Donut Chocolate consistently record the highest demand levels and show a clear upward trend over time. Several other SKUs, including KrimStix Milk and Rebisco Extreme Choco, also exhibit steady growth, though at lower absolute volumes.

In contrast, products such as Whoopie Cheese Variation and Rebisco Crackers Special display relatively low demand across all years, indicating limited contribution to total volume. Overall, the heatmap confirms substantial heterogeneity in demand across SKUs, with growth concentrated in a small number of high-performing products.

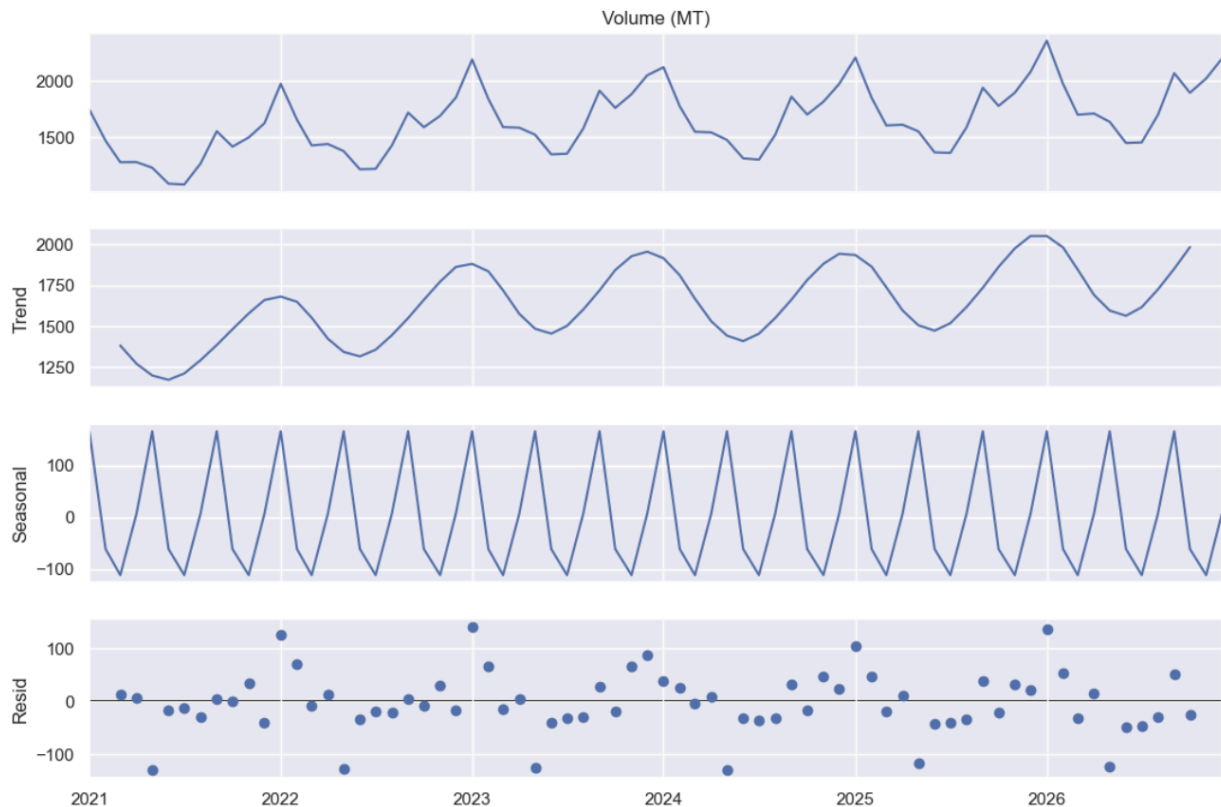


Figure 6: Trend-Seasonal-Residual Decomposition

Figure 6 shows how the time series can be broken down into four major elements: the observed demand (volume), the trend, the seasonal, and the residual elements. In general, the decomposition shows that the trends of total demand are mostly described by a complex of constant upward movement and powerful and stable seasonal trends. The residual part is rather minor and randomly dispersed, which means that the irregular fluctuations or the shocks without explanations have no significant influence. This implies that the underlying demand structure is systematic and predictable and is therefore appropriate to time-series forecasting models which explicitly use trend and seasonality.

Observed component (Volume)

The demand series that is observed has a very apparent cyclical movement in that the peaks and troughs can be traced throughout the whole period. Although it can be seen that short-term movements are present, the overall level of demand is going up over the years, starting with the previous ones, to the subsequent years. This trend indicates an increasing demand situation where seasonality overlays an upward trend. The fact that certain peaks

continue to increase with time also indicates that seasonal demand peaks will only increase with increased overall demand.

Trend component

The trend element shows a steady and constantly increasing trend over the period of study. This means that the demand has been growing in the long term without being seasonal or volatile. This is gradual in nature, implying that the trend in demand is constant and not sudden, since it would be structural expansion due to long term factors like the growth of the market, population changes, or the growth of consumption capacity. Notably, the lack of drastic changes in the trend helps to assume the stability of the long-run demand.

Seasonal component

The seasonal component depicts a very regular and repetitive pattern, and the magnitude and timing are similar in all years. This proves that there is high seasonality, that is, demand rises and falls in a predictable pattern. The fact that seasonal fluctuations have relatively the same strength shows that the intensity of seasonality does not weaken or strengthen dramatically with time. This high stability in seasonality is especially significant to operational planning, where it is possible to predict high- and low-demand seasons with a high level of accuracy and predictability.

Residual component

The residual element oscillates about zero in a random manner, and no noticeable trend or pattern could be observed. Residual values are concentrated close to a specific range, which means that there are only a few unexpected demand shocks or model-unexplained changes. It means that trend and seasonal terms combined explain most of the systematic variation in the data. The decomposition can therefore be said to be effective, and the rest of the noise is probably due to temporary or irregular factors.

Overall insight

Combined, the results of the decomposition point to the fact that the behavior of demand is extremely well-organized, the motivations of which are more long-term expansion and predictable seasonal changes. The smallness and randomness of the residuals suggest well-behaved time series, which further support the appropriateness of decomposition-based and seasonal forecasting techniques for future demand projection and planning.

Product Name	mean	std	CV
Rebisco Crackers Special	24.17	22.72	0.94
Doowee Donut Chocolate	100.65	60.04	0.60
KrimStix Chocolate	159.87	81.08	0.51
Rebisco Extreme Choco	77.86	38.90	0.50
Whoopie Marshmallow Cake	27.83	13.76	0.49
Doowee Donut Strawberry	78.07	36.93	0.47
Rebisco Extreme Mocha	63.65	24.60	0.39
Rebisco Crackers Plain	69.66	14.40	0.21
KrimStix Milk	173.87	35.41	0.20
KrimStix Combo	29.08	5.88	0.20
KrimStix Peanut Butter	58.13	11.72	0.20
Rebisco Crackers Whole Wheat	25.35	4.67	0.18
Whoopie Cheese Variation	4.39	0.81	0.18
Rebisco Crackers Honey Butter	63.21	11.60	0.18

Doowee Donut Assorted	89.22	16.33	0.18
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Table 1: Descriptive Statistics

The coefficient of variation (CV) analysis highlights differences in demand stability across SKUs. Products such as Rebisco Crackers Special (CV = 0.94) and Dowee Donut Chocolate (CV = 0.60) show relatively high variability, indicating demand that is more volatile but still exhibits some structure, likely driven by seasonality. In contrast, core products such as KrimStix Milk (CV = 0.20), KrimStix Combo (CV = 0.20), and Rebisco Crackers Plain (CV = 0.21) display much more stable demand patterns.

Based on common classification practices, most SKUs fall into Group X (CV < 0.5) or Group Y ($0.5 \leq \text{CV} \leq 1.0$), suggesting that demand is generally predictable rather than highly erratic.

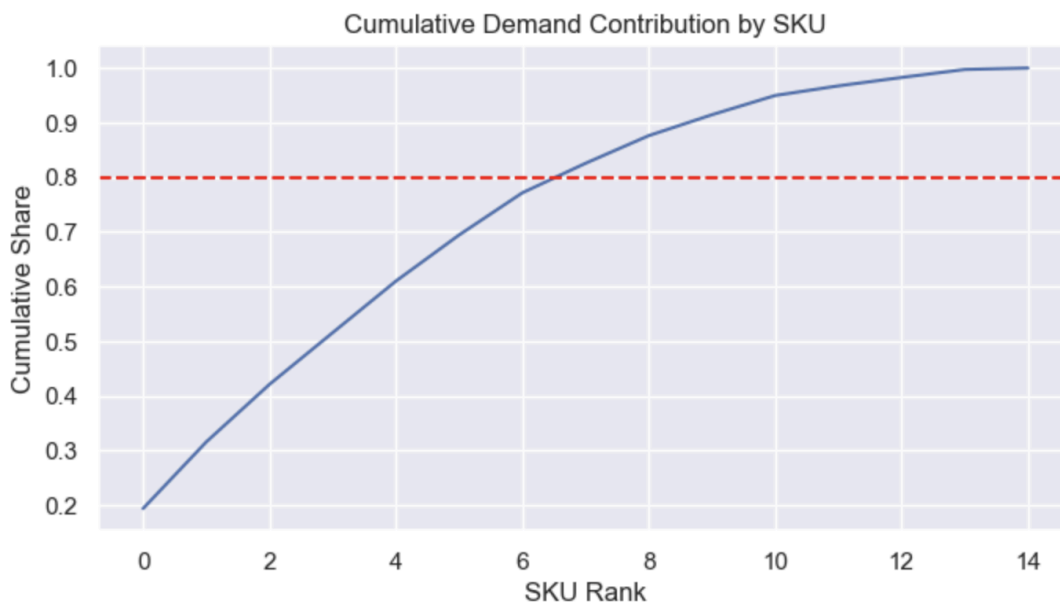


Figure 7: Pareto Chart

The cumulative demand curve indicates that approximately 80% of total demand is generated by the top 6–7 SKUs, clearly confirming a strong Pareto effect. This finding demonstrates that a relatively small group of high-performing products dominates overall demand volume, while the remaining SKUs contribute only marginally to total demand. The steep slope observed in the early part of the curve reflects the disproportionate importance of

these top-ranked SKUs, whereas the flatter tail of the curve suggests diminishing incremental demand contributions from lower-ranked products.

Product Name	
KrimStix Chocolate	23021.43
Doowee Donut Chocolate	14494.19
KrimStix Milk	12518.96
Doowee Donut Strawberry	11242.70
Rebisco Extreme Choco	11211.60
Rebisco Crackers Plain	10031.69
Rebisco Extreme Mocha	9166.13
Doowee Donut Assorted	6423.62
Whoopie Marshmallow Cake	6010.25
Rebisco Crackers Honey Butter	4551.20
KrimStix Peanut Butter	4185.33
KrimStix Combo	2094.12
Rebisco Crackers Whole Wheat	1825.41
Rebisco Crackers Special	1740.23
Whoopie Cheese Variation	316.38
Name: Volume (MT), dtype: float64	

Table 2: SKU Demand Ranking

Table 2 ranks SKUs based on total demand volume and clearly supports the Pareto analysis presented in Figure 7. The results show that demand is highly concentrated among the top 6–7 SKUs, which together account for approximately 80% of total demand.

Specifically, the leading SKUs include KrimStix Chocolate (23,021.43 MT), Doowee Donut Chocolate (14,494.19 MT), KrimStix Milk (12,518.96 MT), Doowee Donut Strawberry (11,242.70 MT), Rebisco Extreme Choco (11,211.60 MT), Rebisco Crackers Plain (10,031.69 MT), and Rebisco Extreme Mocha (9,166.13 MT). These products form the core demand drivers within the portfolio.

In contrast, the remaining SKUs exhibit substantially lower demand volumes and contribute only marginally to overall demand. This pronounced imbalance confirms a strong Pareto effect, where a small subset of high-performing SKUs dominates total demand distribution.

IV. Methodology & Modeling

A rigorous validation framework was implemented to identify the most reliable forecasting "weapon" for Rebisco's operations. Rather than relying on a single approach, we evaluated multiple models to ensure the chosen strategy could withstand market volatility.

In the context of Manufacturing Planning and Control (MPC) systems, our process of using historical data (2021–2025) to train a model and testing it against the known 2026 data before forecasting the future (2027–2032) is fundamentally a **Back-end** operation.

The Back-end: The Technical Engine: According to Jacobs et al. (2011), the Back-end encompasses the execution and technical planning systems. Our current step is classified here because:

Back-testing Methodology: Using a subset of historical data to validate a model's accuracy is a technical evaluation of the "Engine".

Statistical Evaluation: This phase focuses on calculating error metrics such as Mean Absolute Deviation (MAD) or Mean Absolute Percentage Error (MAPE) to ensure the model fits the data.

Data Grounding: We are working with the internal MPC database and historical transactions to optimize the mathematical "truth" before projecting future needs.

1. Data Partitioning Strategy

To verify accuracy before applying the forecast to the long-term 2027–2032 horizon, the dataset was partitioned into two phases

Training Set (2021–2025): This 5-year historical base was used to "teach" the models the underlying growth trends and cyclical patterns of the 15 SKUs.

Testing Set (2026): The 12 months of 2026 were held back as a "real-world exam". Models generated a forecast for this period, and the results were compared against actual values to calculate error metrics (such as MAPE), ensuring the model is reliable for strategic planning.

2. Evaluated Forecasting Models

To forecast demand, we tested several models. These ranged from very simple baselines to advanced methods that capture seasonality and trends. Each model helps us understand sales patterns in different ways.

2.1. Baseline Models (Naive & Moving Average)

Naive Model: This method assumes that next month's sales will be the same as this month's.

Formula:

$$\hat{Y}_{t+1} = Y_t$$

Figure 8: Naive Model formula (Kapil, 2014)

$\hat{y}(t+1)$: forecast for the next period

$y(t)$: actual value in the current period

Moving Average: The forecast is the average of the most recent n months. This reduces short-term fluctuations and makes the overall trend clearer. It is easy to calculate but does not capture seasonality.

Formula:


$$\text{Simple Moving Average} = \frac{(A_1 + A_2 + \dots + A_n)}{n}$$

Figure 9: Simple Moving Average formula (Gajendrakar, 2022)

$A_1, A_2, \dots, A_n$ = actual sales values from the last n periods (e.g., months)

n = number of periods included in the average (for example, 6 months)

2.2. Exponential Smoothing Models

Simple Exponential Smoothing (SES): SES assigns more weight to recent observations and less weight to older ones. It is appropriate when demand is relatively stable and does not exhibit trend or seasonality.

Formula:

$$y_0 = x_0$$

$$y_t = \alpha x_t + (1 - \alpha)y_{t-1}$$

Figure 10: Simple exponential smoothing formula (Radečić, 2021)

x_0 : actual value at time 0

y_0 : initial forecast (set equal to the first actual value)

y_t : forecast at time t

x_t : actual value at time t

α : smoothing constant ($0 < \alpha < 1$)

Holt's Linear Trend Method: Holt's method extends SES by adding a trend component. It estimates both the current level (ℓ) and the rate of change (b), updating them with smoothing constants α and β . The forecast is then projected forward by combining the level and the trend. This makes Holt's method effective for products in a growth or decline phase, where demand is not stable but shows a consistent upward or downward movement.

Formula:

Level:

$$l_t = (1 - \alpha)l_{t-1} + \alpha x_t$$

Trend:

$$b_t = (1 - \beta)b_{t-1} + \beta(l_t - l_{t-1})$$

Model:

$$y_t = l_t + b_t$$

Forecasting:

$$\hat{y}_{t+n} = l_t + nb_t$$

Figure 11: Double exponential smoothing formula (Radečić, 2021)

α : level smoothing constant

x_t : actual value at time t

β : trend smoothing constant

n : number of periods ahead

Level (l_t)

Trend (b_t)

Model (y_t)

Forecast (y^{t+n})

2.3. Holt-Winters Method (Triple Exponential Smoothing)

The Holt-Winters method incorporates seasonality in addition to level and trend. It estimates the baseline level, the trend, and a seasonal factor that repeats every m periods (such as 12 months). By combining these three components, Holt-Winters can anticipate recurring demand spikes, such as those during the Lunar New Year, making it a critical tool for supply chain and procurement planning.

Formula:

l_t : level at time t

α : level smoothing constant ($0 < \alpha < 1$)

x_t : actual value at time t

b_t : trend at time t

β : trend smoothing constant ($0 < \beta < 1$)

c_t : seasonal component at time t

γ : seasonal smoothing constant ($0 < \gamma < 1$)

L : length of seasonality

y_t : estimated value at time t

y^{t+n} : forecast for n periods ahead

$$\begin{aligned}
 &\textbf{Level:} \\
 &l_t = (1 - \alpha)l_{t-1} + \alpha x_t \\
 &\textbf{Trend:} \\
 &b_t = (1 - \beta)b_{t-1} + \beta(l_t - l_{t-1}) \\
 &\textbf{Seasonal:} \\
 &c_t = (1 - \gamma)c_{t-L} + \gamma(x_t - l_{t-1} - b_{t-1}) \\
 &\textbf{Model:} \\
 &y_t = (l_t + b_t)c_t \\
 &\textbf{Forecasting:} \\
 &\hat{y}_{t+n} = (l_t + nb_t)c_{t-L+1+(n-1)modL}
 \end{aligned}$$

Figure 12: Triple exponential smoothing formula (Radečić, 2021)

2.4. Regression with Seasonal Dummies

This regression model captures both trend and seasonality by including dummy variables for each month. It is useful when specific months consistently show higher or lower demand.

2.5. SARIMAX (Seasonal ARIMA with Exogenous Variables)

SARIMAX is a comprehensive forecasting model that integrates autoregression, moving averages, differencing, and seasonal cycles. It also allows the inclusion of exogenous variables, such as promotions or economic indicators, which influence demand. By combining these elements, SARIMAX provides high-precision forecasts and is particularly valuable for long-term strategic planning, such as the Vision 2026–2032 roadmap, where complex demand patterns must be anticipated.

Formula:

The SARIMAX model includes an exogenous term X_t :

$$\phi_p(L)\Phi_P(L^s)(1-L)^d(1-L^s)^D y_t = \theta_q(L)\Theta_Q(L^s)\varepsilon_t + \beta X_t$$

Figure 13: SARIMAX formula (Maharana, 2024)

p: number of AR terms

L: lag operator

P: number of seasonal AR terms

s: seasonal period

d: degree of non-seasonal differencing

D: degree of seasonal differencing

q: number of MA terms

Q: number of seasonal MA terms

Random noise or residuals at time t

V. Model Evaluation and Selection

1. Analyse the chart

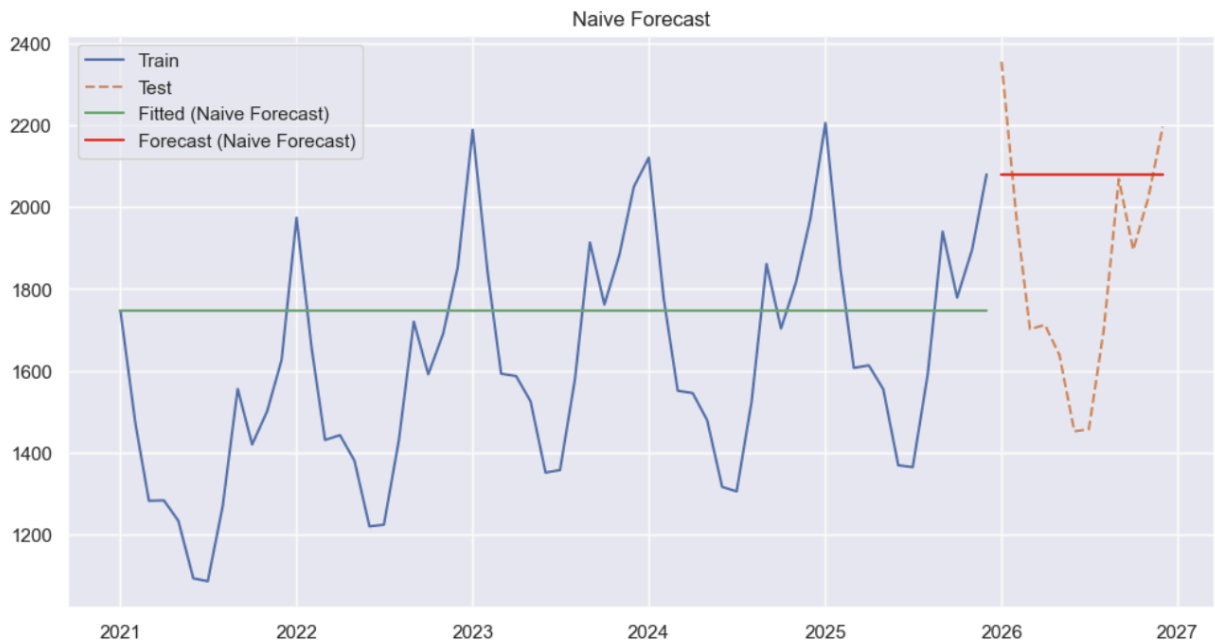


Figure 14:Naive Forecast Method Result

The Naive model was trained on data from 2021 to 2025 and used to forecast demand for 2026. The fitted values do not align with the training data because the model simply uses the last observed value as the forecast, resulting in a flat fitted line that ignores fluctuations. In the test period, the forecast remains constant and fails to capture the strong seasonal variations observed in actual demand. This mismatch clearly indicates that the Naive model is unsuitable for forecasting demand when trend and seasonality are present.

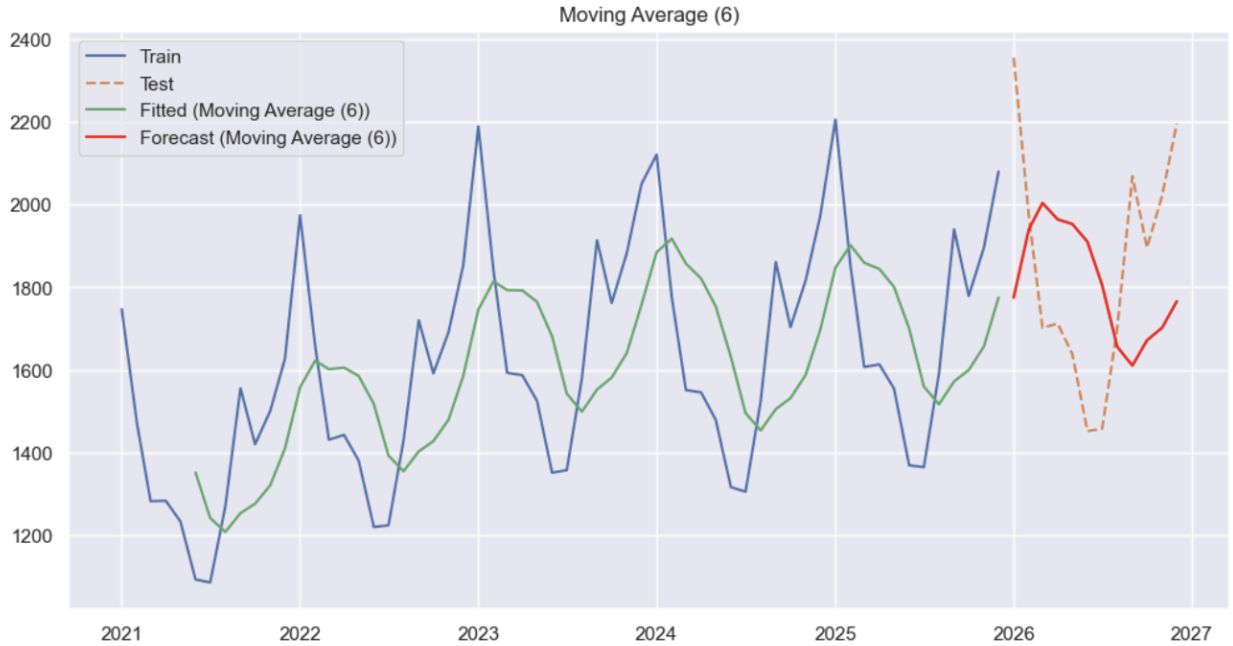


Figure 15: Moving Average (6) Forecast Result

The Moving Average (6) model, where demand is calculated as the average of the previous six months, was trained on data from 2021 to 2025 and used to forecast demand for 2026. The fitted values smooth the training data and capture the general demand level, but they do not match the actual series during periods of sharp increases or decreases due to the lag introduced by averaging. In the test period, the forecast partially follows the direction of demand but fails to align with the timing and magnitude of seasonal fluctuations. This mismatch indicates that while the Moving Average (6) model reduces noise, it is not suitable for accurate forecasting when strong seasonality persists.

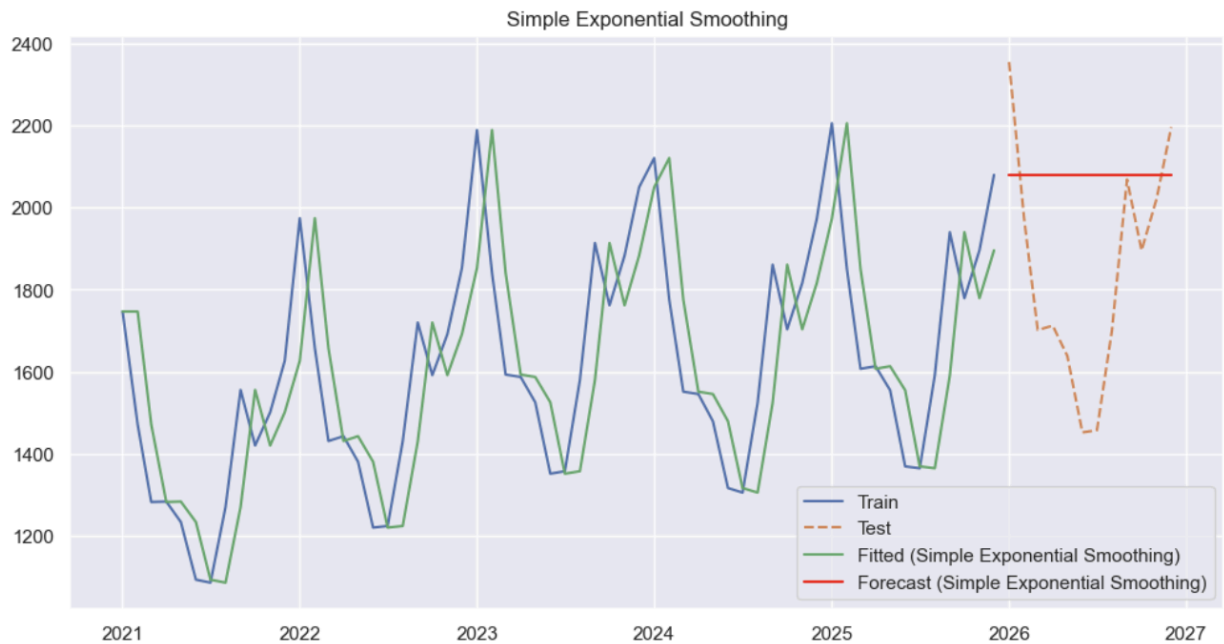


Figure 16: Simple Exponential Smoothing (SES) Forecast Result

The SES model was trained on data from 2021 to 2025 and used to forecast demand for 2026. Although the fitted values follow the general level of the training data, the forecast for 2026 is flat due to the model's assumption of stable demand. The test data in 2025–2026 shows strong seasonal fluctuations, which are not captured by the forecast line. This mismatch indicates that SES fails to model seasonality, leading to poor alignment between the test observations and the forecast. As a result, SES is not suitable for accurately forecasting demand when seasonal variation remains significant.

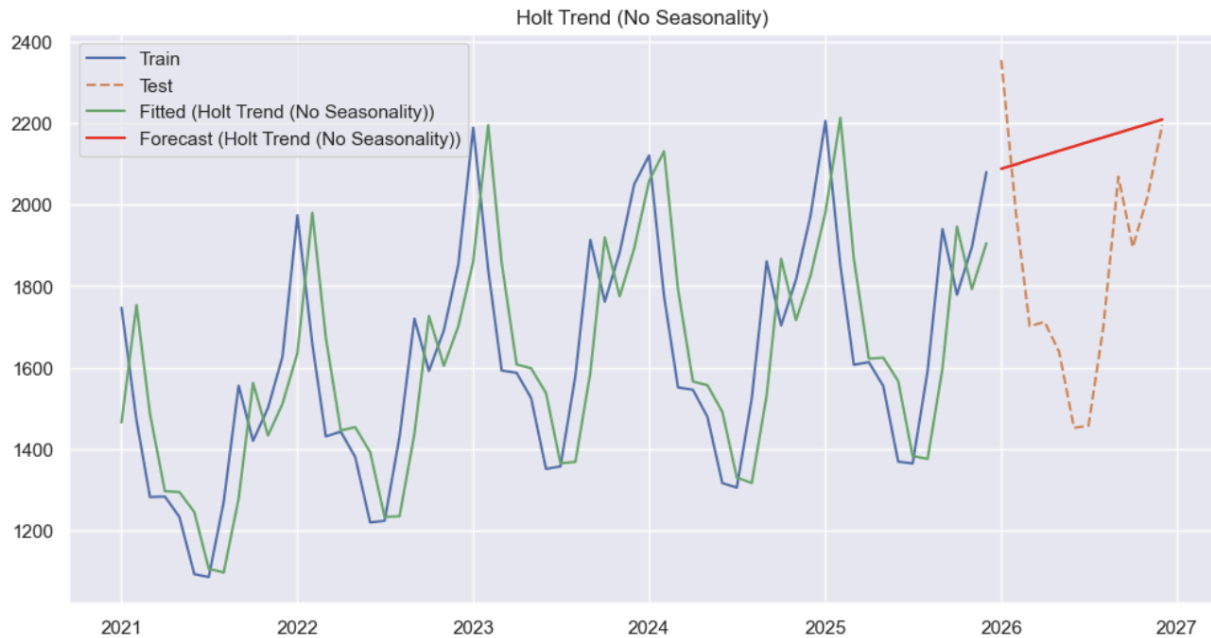


Figure 17: Holt Trend Model (No Seasonality) Forecast Result

The Holt Trend model was trained on data from 2021 to 2025 and used to forecast demand for 2026. The fitted values track the overall upward trend of the training data reasonably well, indicating that the model successfully captures the long-term growth in demand. However, because the model does not include a seasonal component, it fails to reflect the recurring peaks and troughs observed in the data. As a result, the forecast for 2026 follows a smooth upward trajectory that does not align with the test data, which continues to exhibit strong seasonal fluctuations. This mismatch shows that while the Holt model captures trend effectively, it is inadequate when seasonality remains significant.

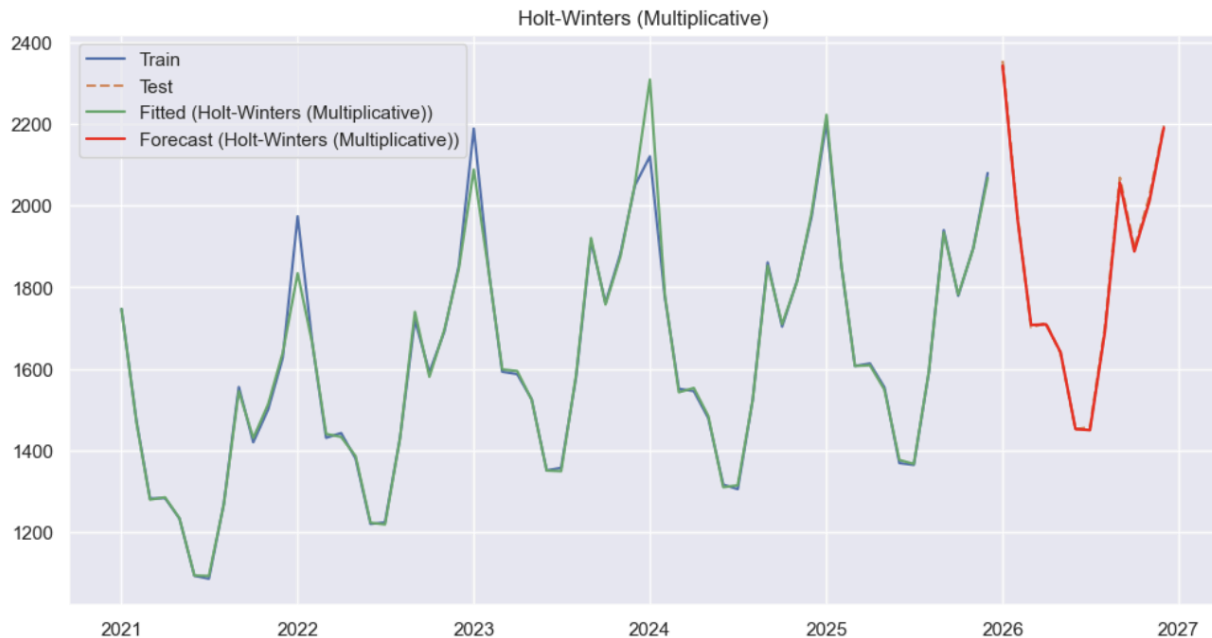


Figure 18: Holt–Winters (Multiplicative) Forecast Result

The Holt–Winters (Multiplicative) model was trained on data from 2021 to 2025 and used to forecast demand for 2026. The fitted values closely follow the training data, capturing both the upward trend and the strong seasonal fluctuations present in the series. This indicates that the model effectively learns the interaction between demand level and seasonality. When compared with the test data, the forecast for 2026 reflects similar seasonal patterns and magnitude of variation, although some deviations remain. Overall, the Holt–Winters (Multiplicative) model provides the most realistic forecast among the evaluated methods, making it suitable for forecasting demand under persistent seasonality.

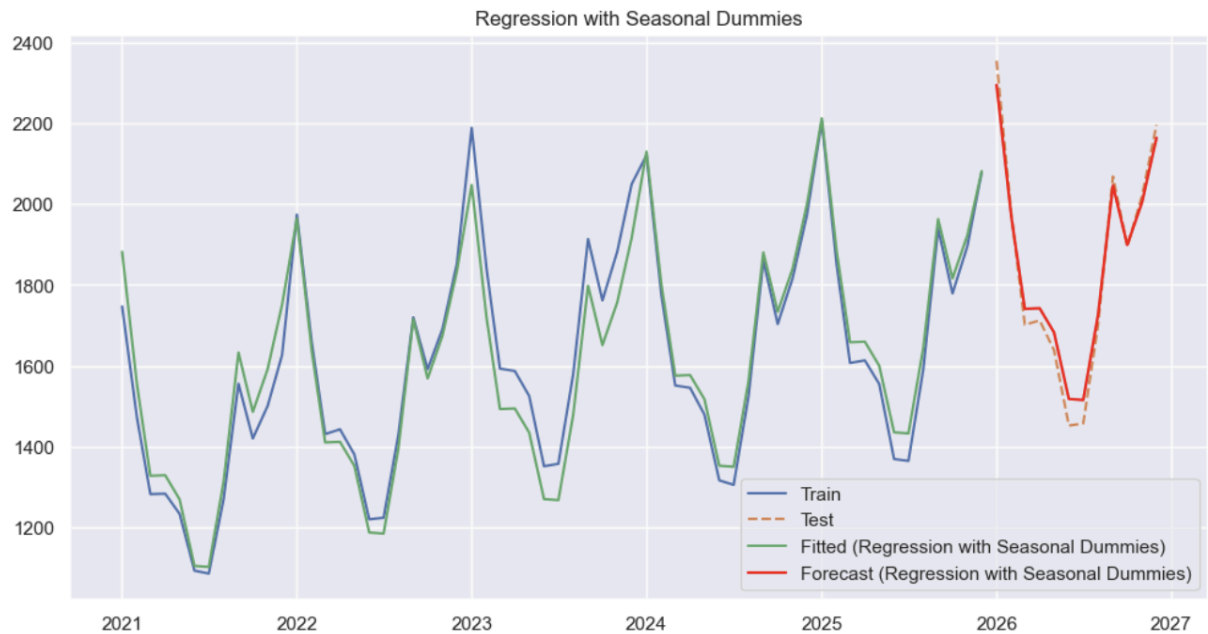


Figure 19: Regression with Seasonal Dummies Forecast Result

The regression model with seasonal dummy variables was trained on data from 2021 to 2025 and used to forecast demand for 2026. Seasonal dummy variables are binary indicators that represent different seasonal periods and allow the model to capture recurring seasonal effects by adjusting the demand level for each season relative to a baseline period. Although the fitted values capture the general seasonal pattern in the training data, they do not fully match the actual observations, especially at peak and trough periods. In the test period, the forecast does not align well with the observed demand, showing noticeable deviations in both timing and magnitude. This mismatch indicates that while the model accounts for seasonality in a simplified and fixed form, it lacks the flexibility to adapt to changes in seasonal intensity over time. Therefore, the regression with seasonal dummies provides limited accuracy for forecasting demand under strong and varying seasonality.

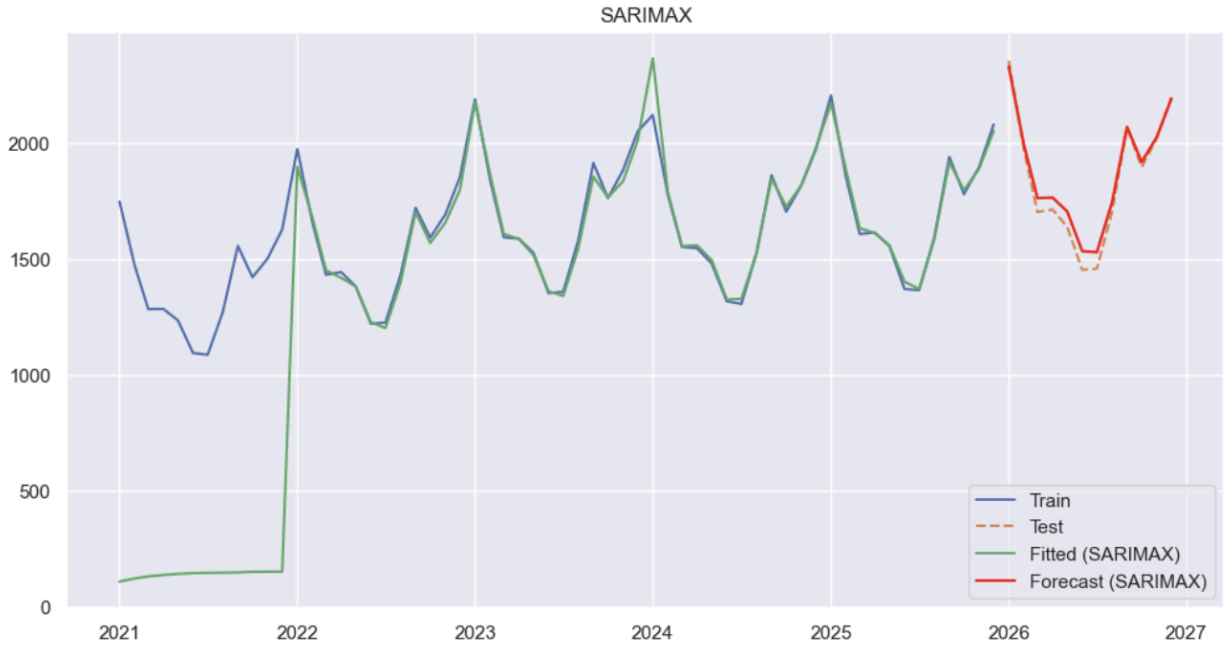


Figure 20: SARIMAX Forecast Result

The SARIMAX model was trained on data from 2021 to 2025 and used to forecast demand for 2026. After the initial periods, the fitted values align relatively well with the training data, capturing both the overall trend and recurring seasonal fluctuations. However, some instability is observed at the beginning of the fitted series, indicating sensitivity to model initialization. In the test period, the forecast follows the general seasonal pattern but does not fully match the observed demand, with noticeable deviations in magnitude. This suggests that while SARIMAX is capable of modeling complex temporal structures, its forecasting accuracy depends strongly on model specification and parameter estimation.

2. Model Evaluation

To support Rebisco's Vision 2026-2032, we tested seven forecasting models using the 2026 testing test. We compared them using five important metrics:

MAPE (Mean Absolute Percentage Error): Shows how accurate the forecast is, in percentage.

MAD (Mean Absolute Deviation): Measures how far forecasts are from actual sales, on average.

MSE and RMSE: These show how large the errors are, especially when they are far off.

Bias: Tells us if the model tends to over-forecast or under-forecast.

	Model	MAD	MSE	RMSE	MAPE (%)	Bias
0	Naive	296.938333	127693.395967	357.342127	17.837413	-231.733333
1	Moving Average (6)	313.404583	122341.957041	349.774151	17.098349	34.593750
2	SES	296.938332	127693.394697	357.342126	17.837413	-231.733331
3	Holt	345.013229	163931.062561	404.884011	20.668183	-300.578726
4	Holt-Winters (Mul)	7.119545	65.646816	8.102272	0.373863	5.663803
5	Regression + Seasonal Dummy	33.595083	1512.259266	38.887778	1.929741	-10.790500
6	SARIMAX	37.453027	2128.630144	46.137080	2.268643	-32.338819

Table 3: Forecast Accuracy Comparison

Among these models, the Holt-Winters method stood out as the most accurate, achieving an exceptionally low MAPE of 0.37 percent. This result is far superior to baseline models such as the Naïve Forecast, which recorded a MAPE of 17.83 percent, and the Moving Average model, which had 17.09 percent. Holt-Winters also delivered the lowest MAD, MSE, and RMSE values, confirming its reliability and consistency. Importantly, this model is well-suited to Rebisco’s product portfolio because it captures both growth trends and complex monthly seasonality, allowing the supply chain team to anticipate demand spikes such as those during Lunar New Year and adjust procurement lead times accordingly.

Bias is another critical measure in supply chain forecasting because it shows whether a model tends to over-forecast or under-forecast. A negative bias means forecasts are higher than actual demand, which can lead to excess inventory, higher storage costs, and risks of product expiration. A positive bias means forecasts are lower than actual demand, which increases the risk of stock-outs, lost sales, and customer dissatisfaction. The ideal Bias is close to zero, indicating a balanced model. Even when a model has a low MAPE, a strong Bias in one direction can still create risks. For example, SARIMAX achieved a competitive MAPE of 0.42 percent but showed a negative Bias, suggesting a tendency to over-forecast. In such cases, production planning would need to be adjusted downward to avoid surplus.

Overall, Holt-Winters (Multiplicative) is the recommended forecasting model for Rebisco’s Vision 2026-2032 roadmap. It provides the best balance of accuracy, reliability, and

operational safety, outperforming all other models across the key metrics. By combining low error rates with a near-zero Bias, Holt-Winters ensures forecasts are both precise and practical, making it the most reliable foundation for long-term supply chain planning.

3. Selection and Final Demand Forecasting

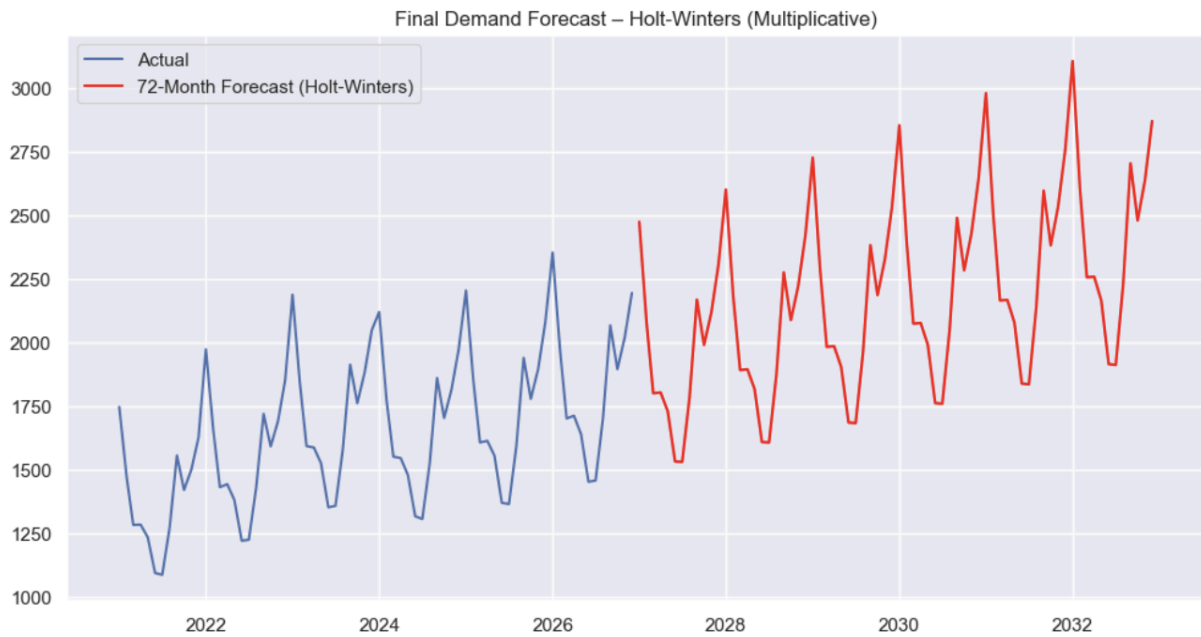


Figure 21: Holt–Winters Multiplicative Forecast Result

Based on the comparative analysis of seven forecasting models, the Holt–Winters (Multiplicative) model is selected as the most appropriate method for forecasting material demand. The model demonstrates the best fit to historical data from 2021 to 2026 by effectively capturing both the upward trend and strong seasonal fluctuations observed in the series.

When forecasting demand for 2026, the Holt–Winters (Multiplicative) model produces predictions that closely align with the test data in terms of seasonal pattern and magnitude, outperforming models that ignore seasonality or oversmooth the data. This indicates that the multiplicative seasonal structure is suitable for demand data where seasonal variation changes proportionally with the demand level.

Given the assumption that demand behavior observed between 2025 and 2026 remains stable, the Holt–Winters (Multiplicative) model is considered reliable for extending the forecast

horizon to 2032. Therefore, this model is adopted as the final forecasting approach for long-term material planning.

Data forecasting from 2027 to 2032, getting the data [here](#).

VI. Supply Chain & Logistics Implications

The selection of the Holt–Winters (Multiplicative) model for demand forecasting has several important implications for supply chain and logistics planning. Since the model effectively captures both trend and strong seasonality, it enables firms to anticipate fluctuations in material demand more accurately over the long term.

First, improved demand visibility allows procurement teams to plan raw material purchasing in advance, reducing the risk of stockouts during peak demand periods and excess inventory during low seasons. This supports more efficient inventory management and lowers holding and emergency procurement costs.

Second, accurate seasonal forecasts enhance production planning by allowing manufacturers to align capacity, labor scheduling, and production cycles with expected demand patterns. This reduces operational disruptions and improves capacity utilization.

Third, logistics and transportation planning can be optimized based on forecasted demand peaks and troughs. Firms can proactively arrange transportation capacity, negotiate long-term contracts with logistics providers, and avoid last-minute shipments that often increase logistics costs.

Finally, the assumption that demand patterns observed between 2026 and 2027 remain stable until 2032 supports strategic supply chain decisions such as warehouse capacity planning, supplier relationship management, and long-term logistics network design. Overall, adopting the Holt–Winters (Multiplicative) model strengthens supply chain resilience and supports cost-effective, demand-driven logistics operations.

VII. Conclusion

The use of advanced time-series forecasting has given Rebisco Vietnam a strong advantage in managing its supply chain. By applying the Holt-Winters (Multiplicative) model,

the company can plan ahead instead of only reacting to changes. The forecast shows demand will rise steadily from about 2,000 metric tons in 2027 to nearly 3,000 metric tons by 2032, while still reflecting seasonal peaks each year. This reliable forecast helps Rebisco avoid stock-outs and excess inventory, making the Vision 2026-2032 not only a growth plan but also a practical and financially safe roadmap for the future.

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